3/10/2026

Practical 11

a) Simulation of Static Routing Configuration using CISCO Packet Tracer

Aim: To simulate static routing configuration using cisco packet tracer.

Static Routing:

Static routes are the routes manually added to the router's routing table. The process of adding static routes to the routing table is known as static routing.

If two routes are specified for the same destination the route with the lower Administrative Distance (AD) is selected as the main route. A route with a higher AD can be configured as a backup route. Routers automatically learn their directly connected networks. Static routes are required only for networks that are not directly connected to the router

Connected and Remote Networks

Router	Networks on local interfaces	Networks on other routers
Router 0	10.0.0.0/8, 20.0.0.0/8, 40.0.0.0/8	30.0.0.0/8, 50.0.0.0/8
Router 1	20.0.0.0/8, 30.0.0.0/8, 50.0.0.0/8	10.0.0.0/8, 40.0.0.0/8
Router 2	40.0.0.0/8, 50.0.0.0/8	10.0.0.0/8, 20.0.0.0/8, 30.0.0.0/8

Router 0 Configuration

Requirements:

* Create two routes for network 30.0.0.0/8. Router 1 is the main route and router 2 is the backup.

* Create two routes for host 30.0.0.100/32. Router 2 is the main route and router 1 is the backup
* Create two routes for network 50.0.0.0/8. Router 2 is the main route and Router 1 is the backup.
* Verify the routing table.

Router 1 Configuration:

Requirements:

* Create two routes for network 10.0.0.0/8. Router 0 is the main route and the other route is the backup.
* Create two routes for network 40.0.0.0/8. Router 0 is the main route and Router 2 is the backup.
* Verify the routing table.

Router 2 Configuration:

* Create static routes for network 10.0.0.0/8 and 30.0.0.0/8 and verify the routes.

Verify static routing:

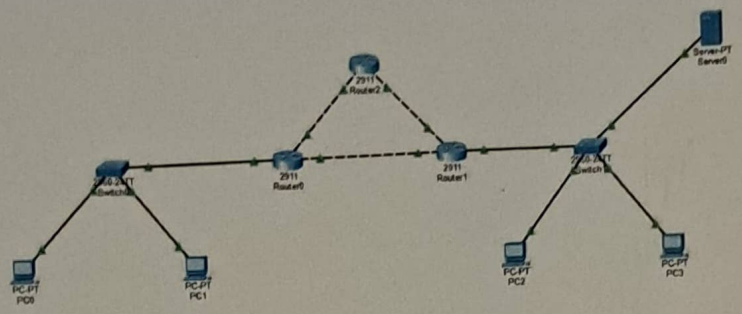
* Use show ip route static to display the static routes in the routing table.
* Use ping to verify connectivity between PCs
* Use tracert from a PC to check the path followed by packets.
* The main route has a lower AD value and is used first.
* To verify the backup route, disconnect/delete the link between Router 0 and Router 1 and check the connectivity again.

Deleting a static route

To delete a static route:

* Use show ip route static to display the static routes.
* Note down the route to be deleted.
* Use the no ip route command to delete the route

If a backup route is available, it becomes the main route when the main route is deleted.




```
Router0#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.1        YES manual up                    up
GigabitEthernet0/1     20.0.0.1        YES manual up                    up
GigabitEthernet0/2     40.0.0.1        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

```
Router1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     30.0.0.1        YES manual up                    up
GigabitEthernet0/1     20.0.0.2        YES manual up                    up
GigabitEthernet0/2     50.0.0.2        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

```
Router2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     40.0.0.2        YES manual up                    up
GigabitEthernet0/1     50.0.0.1        YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

```
Router0#show ip route static
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S       30.0.0.0/8 [10/0] via 20.0.0.2
S       30.0.0.100/32 [10/0] via 40.0.0.2
S    50.0.0.0/8 [10/0] via 40.0.0.2
```

```
Router1#show ip route static
S    10.0.0.0/8 [10/0] via 20.0.0.1
S    40.0.0.0/8 [10/0] via 20.0.0.1
```

```
Router2#show ip route static
S    10.0.0.0/8 [1/0] via 40.0.0.1
S    30.0.0.0/8 [1/0] via 50.0.0.2
```

```
C:\>tracert 30.0.0.20

Tracing route to 30.0.0.20 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    0 ms      0 ms      0 ms      20.0.0.2
  3    0 ms      0 ms      0 ms      30.0.0.20

Trace complete.
```

```
C:\>tracert 30.0.0.100

Tracing route to 30.0.0.100 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    0 ms      0 ms      0 ms      40.0.0.2
  3    0 ms      0 ms      0 ms      20.0.0.2
  4    0 ms      0 ms      0 ms      30.0.0.100

Trace complete.
```

```
Router0>enable
Router0#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router0(config)#interface gigabitEthernet0/1
Router0(config-if)#shutdown

Router0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

Router0(config-if)#exit
Router0(config)#exit
```

```
Router0#show ip route static
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S       30.0.0.0/8 [20/0] via 40.0.0.2
S       30.0.0.100/32 [10/0] via 40.0.0.2
S    50.0.0.0/8 [10/0] via 40.0.0.2
```

```
C:\>tracert 30.0.0.20

Tracing route to 30.0.0.20 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    0 ms      0 ms      0 ms      40.0.0.2
  3    0 ms      0 ms      0 ms      50.0.0.2
  4    0 ms      0 ms      0 ms      30.0.0.20

Trace complete.
```

```
Router0#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router0(config)#interface gigabitEthernet0/1
Router0(config-if)#no shutdown

Router0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router0(config-if)#exit
Router0(config)#exit
```

```
Router0#show ip route static
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S       30.0.0.0/8 [10/0] via 20.0.0.2
S       30.0.0.100/32 [10/0] via 40.0.0.2
S    50.0.0.0/8 [10/0] via 40.0.0.2
```

```
Router0#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router0(config)#no ip route 30.0.0.100 255.255.255.255 40.0.0.2
Router0(config)#exit
Router0#
%SYS-5-CONFIG_I: Configured from console by console

Router0#show ip route static
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S       30.0.0.0/8 [10/0] via 20.0.0.2
S       30.0.0.100/32 [20/0] via 20.0.0.2
S    50.0.0.0/8 [10/0] via 40.0.0.2

Router0#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router0(config)#no ip route 30.0.0.100 255.255.255.255 20.0.0.2
Router0(config)#exit
Router0#
%SYS-5-CONFIG_I: Configured from console by console

Router0#show ip route static
S    30.0.0.0/8 [10/0] via 20.0.0.2
S    50.0.0.0/8 [10/0] via 40.0.0.2
```

Result :
Thus, static routing was configured and verified successfully using Cisco Packet Tracer.

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Practical 11

b) Simulate RIP using CISCO Packet Tracer

Aim: To simulate RIP using CISCO Packet Tracer

Initial IP configuration

Device	Interface	IP configuration	Connected with
PC0	Fast Ethernet	10.0.0.2/8	Router0's Fa0/1
Router 0	Fa0/1	10.0.0.1/8	PC0's Fast Ethernet
Router0	S0/0/1	192.168.1.254/30	Router2's S0/0/1
Router0	S0/0/0	192.168.1.249/30	Router1's S0/0/1
Router1	S0/0/0	192.168.1.250/30	Router0's S0/0/0
Router1	S0/0/1	192.168.1.246/30	Router2's S0/0/0
~~Router1~~	~~S0/0~~	192.168.245/30	Router1's S0/0/1
Router2	S0/0/0	192.168.1.253/30	Router0's S0/0/1
Router2	S0/0/1	20.0.0.1/30	PC1's Fast Ethernet
Router2	Fa0/1	20.0.0.2/30	Router2's Fa0/1
PC1	Fast Ethernet		

Assign IP addresses to PCs

Double click the PCs → Desktop → IP configuration and assign the IP addresses according to the above table.

Assign IP addresses to interfaces of routers

Configure the IP addresses of the router interfaces according to the given IP configuration table.

For serial interfaces, clock rate and bandwidth are configured only at the DCE end. The no shutdown command is used to bring the interface up.

Configure RIP routing protocol:

RIP routing configuration requires two steps:

* Enable RIP routing protocol from global configuration mode.
* Specify the directly connected networks to be advertised.

RIP automatically exchanges routing information between the routers.

Router0

Networks to be advertised:

10.0.0.0

192.168.1.252

192.168.1.248

Router1

Networks to be advertised:

192.168.1.244

192.168.1.248.

Router2

Networks to be advertised:

20.0.0.0

192.168.1.252

192.168.1.244

Verification

The network can be verified using the ping command.

RIP automatically manages the routes. If one route goes down, RIP automatically switches to another available route.

Route 1:

PC0 → Router 0 → Router 2 → PC1

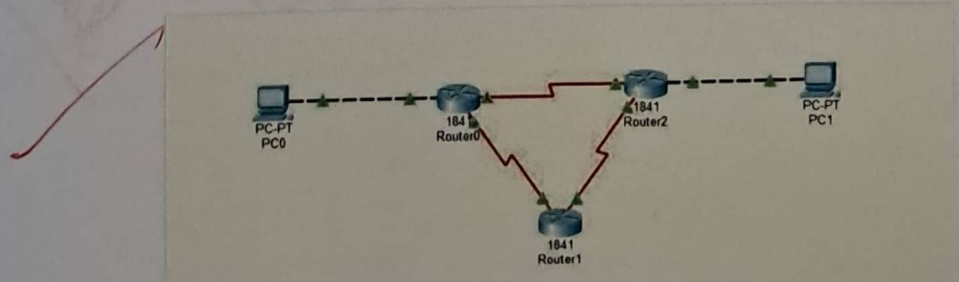
Route 2:

PC0 → Router 0 → Router 1 → Router 2 → PC1

By default, RIP uses the route with the lowest hop count. Route 1 has fewer hops and is therefore selected.

The tracert command can be used to verify the route.

If route 1 fails disconnect the cable between Router 0 S0/0/1 and Router 2 S0/0/1. RIP automatically reroutes the traffic through Route 2. Use tracert again to verify the new route.



```
Router0#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES manual administratively down down
FastEthernet0/1        10.0.0.1        YES manual up                    up
Serial0/1/0            192.168.1.249   YES manual up                    up
Serial0/1/1            192.168.1.254   YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

```
Router1>enable
Router1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/1/0            192.168.1.250   YES manual up                    up
Serial0/1/1            192.168.1.246   YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

```
Router2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        20.0.0.1        YES manual up                    up
FastEthernet0/1        unassigned      YES manual administratively down down
Serial0/1/0            192.168.1.245   YES manual up                    up
Serial0/1/1            192.168.1.253   YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

```
Router0#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

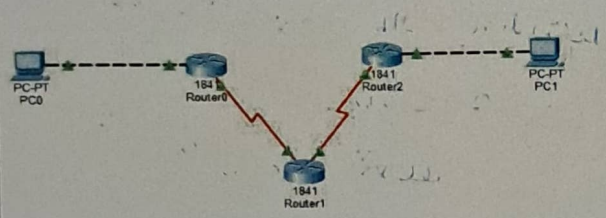
C    10.0.0.0/8 is directly connected, FastEthernet0/1
R    20.0.0.0/8 [120/1] via 192.168.1.253, 00:00:09, Serial0/1/1
     192.168.1.0/30 is subnetted, 3 subnets
R       192.168.1.244 [120/1] via 192.168.1.250, 00:00:23, Serial0/1/0
                      [120/1] via 192.168.1.253, 00:00:09, Serial0/1/1
C       192.168.1.248 is directly connected, Serial0/1/0
C       192.168.1.252 is directly connected, Serial0/1/1
```

```
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time=51ms TTL=126
Reply from 10.0.0.2: bytes=32 time=27ms TTL=126
Reply from 10.0.0.2: bytes=32 time=24ms TTL=126
Reply from 10.0.0.2: bytes=32 time=29ms TTL=126

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 24ms, Maximum = 51ms, Average = 32ms
```




```
Router0>enable
Router0#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/1
R    20.0.0.0/8 [120/2] via 192.168.1.250, 00:00:02, Serial0/1/0
     192.168.1.0/30 is subnetted, 2 subnets
R       192.168.1.244 [120/1] via 192.168.1.250, 00:00:02, Serial0/1/0
C       192.168.1.248 is directly connected, Serial0/1/0
```

```
C:\>tracert 10.0.0.2

Tracing route to 10.0.0.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      20.0.0.1
  2    12 ms     20 ms     19 ms     192.168.1.246
  3    29 ms     37 ms     32 ms     192.168.1.249
  4    14 ms     24 ms     26 ms     10.0.0.2

Trace complete.
```

Result:
Thus, RIP was successfully simulated and verified using Cisco Packet Tracer.